**1. Execute the following commands**

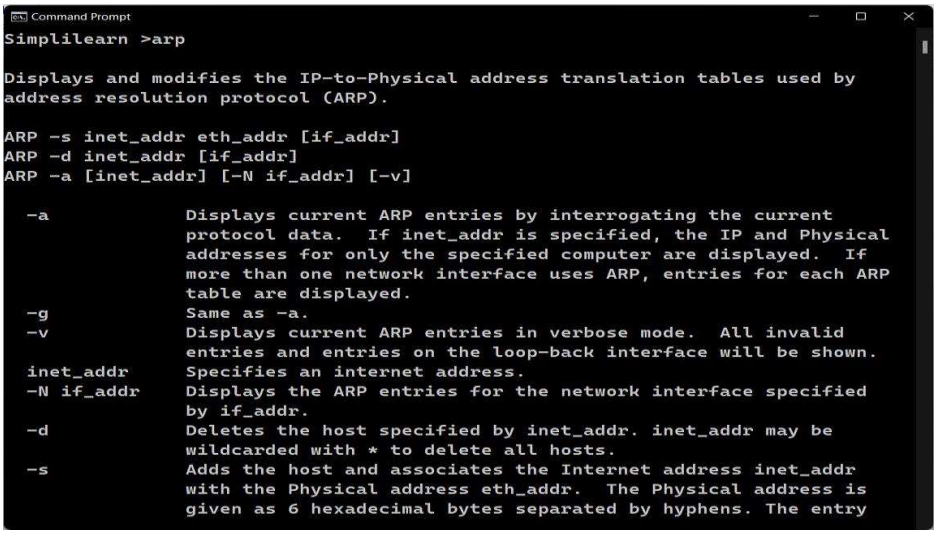
**Arp, ipconfig, hostname, netdiag, netstat, nslookup, pathping, pingroute, tracert**.

Sol: The Windows operating system provides its user with a powerful tool, i.e., Command Prompt, which allows us to access and configure system settings and data. In this article on 'Networking Commands'. These network commands prove helpful when there is a need to configure or troubleshoot the network settings of our system.

What Are Networking Commands?

Every system is connected to numerous different networks and systems through internal or external network channels. These network settings often run into issues and affect the system's working. Such network problems can be resolved using 'networking commands.

**1.ARP(Address Resolution Protocol):**




The ARP command is used to access the mapping structure of IP addresses to the MAC address. This provides us with a better understanding of the transmission of packets in the network channel.

Command to enter in Prompt - arp

**2. IPCONFIG:**



The IPCONFIG network command provides a comprehensive view of information regarding the IP address configuration of the device we are currently working on. The IPConfig command also provides us with some variation in the primary command that targets specific system settings or data, which are:

- **IPConfig/all** - Provides primary output with additional information about network adapters.
- **IPConfig/renew** - Used to renew the system's IP address.
- **IPConfig/release** - Removes the system's current IP address.

Command to enter in Prompt – ipconfig.

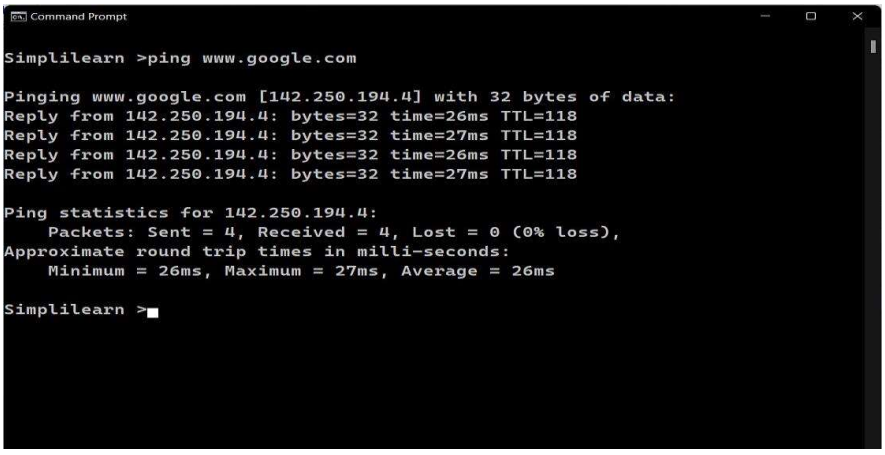
## 3.NSLOOKUP




The NSLOOKUP command is used to troubleshoot network connectivity issues in the system. Using the nslookup command, we can access the information related to our system's DNS server, i.e., domain name and IP address.

Command to enter in Prompt - nslookup

## 4.PING:




The Ping command is one of the most widely used commands in the prompt tool, as it allows the user to check the connectivity of our system to another host.

This command sends four experimental packets to the destination host to check whether it receives them successfully, if so, then, we can communicate with the destination host. But in case the packets have not been received, that means, no communication can be established with the destination host.
Command to enter in Prompt - ping www.destination_host_name.com

**5.HOSTNAME**:




The HOSTNAME command displays the hostname of the system. The hostname command is much easier to use than going into the system settings to search for it.
Command to enter in Prompt – hostname

**6.TRACERT**:




The TRACERT command is used to trace the route during the transmission of the data packet over to the destination host and also provides us with the “hop” count during transmission.
Using the number of hops and the hop IP address, we can troubleshoot network issues and identify the point of the problem during the transmission of the data packet.
Command to enter in Prompt- tracert IP-address OR tracert www.destination_host_name.com

## 7.NETSTAT:

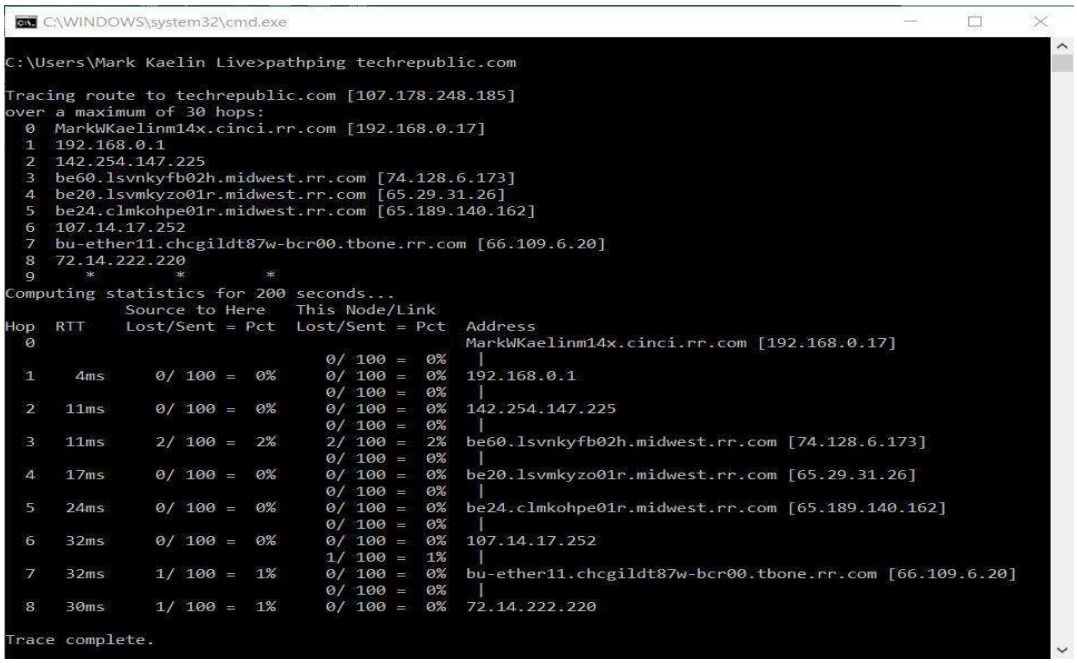



The Netstat command as the name suggests displays an overview of all the network connections in the device. The table shows detail about the connection protocol, address, and the current state of the network. Command to enter in Prompt – netstat

## 8. PATHPING:

"Pathping" is similar to "tracert," except it's more informative and takes a lot longer to execute. After sending out packets from you to a given destination, it analyzes the route taken and computes packet loss on a per-hop basis.

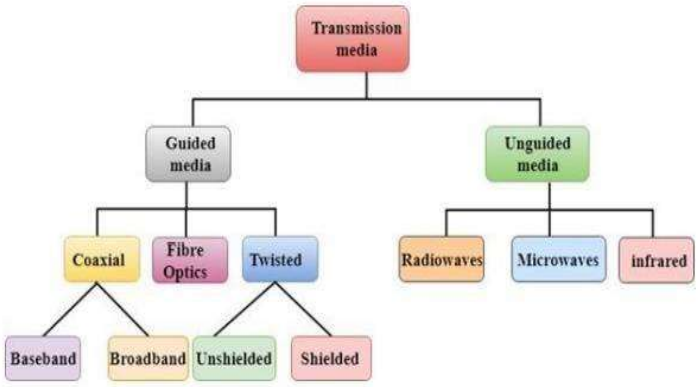
Sample usage and output:



## PROGRAM- 2: STUDY OF DIFFERENT TYPES OF CABLES

### Transmission Medium:

A communication channel that is used to carry the data from one transmitter to the receiver through the electromagnetic signals. The main function of this is to carry the data in the bits form through the Local Area Network(LAN). In data communication, it works like a physical path between the sender & receiver. For instance, in a copper cable network the bits in the form of electrical signals whereas in a fiber network ,the bits are available in the form of light pulses. The quality as well as characteristics of data transmission, can be determined from the characteristics of medium &signal. The properties of different transmission media are delay, bandwidth, maintenance, cost and easy installation.

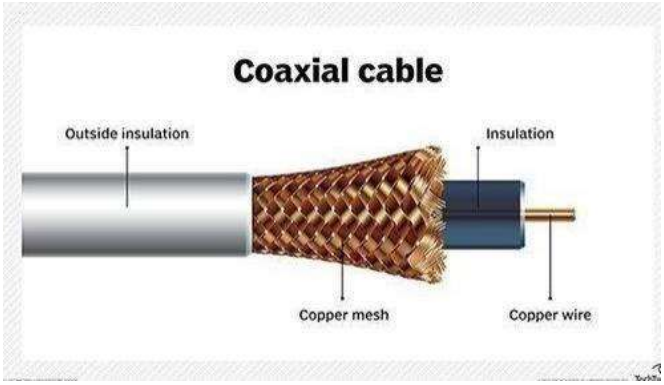



### Bounded/Guided Transmission Media:

This kind of transmission media is also known as wired otherwise bounded media. In this type, the signals can be transmitted directly & restricted in a thin path through physical links. The types of Bounded /Guided transmission are discussed below.

### Coaxial Cable:

Coaxial cable (or coax) carries signals of higher frequency ranges than those in twisted pair cable. It has a central core conductor of solid or stranded wire (usually copper) enclosed in an insulating sheath, which is, in turn, encased in an outer conductor of metal foil, braid, or a combination of the two. The outer conductor is also enclosed in an insulating sheath, and the whole cable is protected by a plastic cover.



**Applications:**

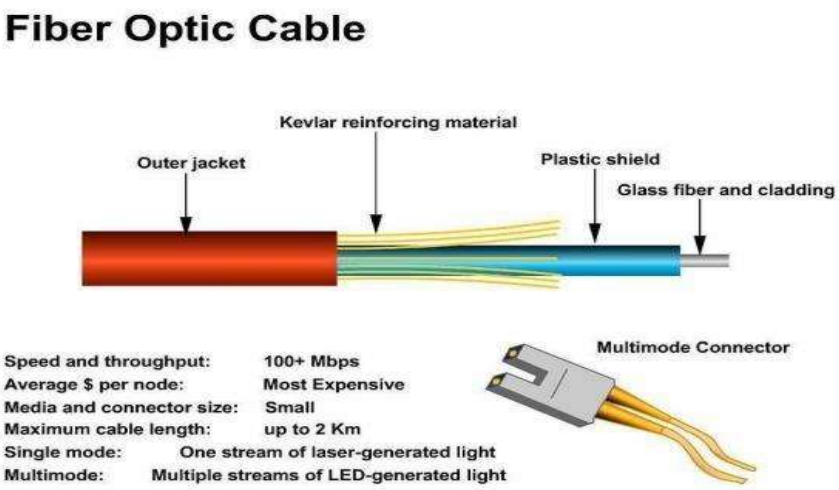
1. Coaxial cable was widely used for both analog and digital data transmissions.
2. It has higher bandwidth.
3. Inexpensive when compared to fiber optical cables.
4. It uses for longer distances at higher data rates.
5. Excellent noise immunity.
6. Used in LAN and Television distribution.
7. **Disadvantage :**

1.Single cable failure can fail the entire network.
2.Difficult to install and expensive when compared with twisted pairs.
3.If the shield is imperfect, it can lead to grounded loop.

**Fibre Optic Cable:**

A fiber optic cable is a network cable that contains strands of glass fibers inside an insulated casing.

They're designed for long-distance, high-performance data networking, and telecommunications. Compared to wired cables, fiber optic cables provide higher bandwidth and transmit data over longer distances. Fiber optic cables support much of the world's internet, cable television, and telephone systems.




**Advantages of Fiber Optic Cables:**

1.The loss of signal in optical fiber is less than that in copper wire.
2.Opticalfibers usually have a longer life cycle for over 100 years.

Disadvantage:

1.It is expensive.
2.Difficult to install.